



WORLD UNIVERSITY OF BANGLADESH

Software Requirements Specification (SRS) on:

National Hospital Management System (NHMS)

Course Title: Software Engineering

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Chapter -01

Introduction:

1.1 Purpose of the System:

The purpose of the National Hospital Management System (NHMS) is to develop a centralized healthcare management platform that can simplify and automate different activities of hospitals. The system is designed to manage patient information, doctor information, appointments, medical records, laboratory reports, billing information, and other hospital-related operations.

The main goal of this system is to reduce manual paperwork, improve data accuracy, provide faster healthcare services, and ensure efficient communication between patients, doctors, and hospital authorities.

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the NHMS. It provides a complete guideline for designing, developing, testing, and maintaining the system.

1.2 Scope of the Project:

The National Hospital Management System will provide a digital platform for managing hospital activities efficiently. The system will allow different users such as administrators, doctors, receptionists, laboratory staff, and patients to perform their required operations according to their roles.

The system will include the following major features:

Patient Management: The system will store and manage patient information including personal details, medical history, and previous visit records.

Doctor Management: The system will maintain doctor profiles, specialization details, availability schedules, and assigned patients.

Appointment Management: Patients can book appointments with available doctors. Receptionists can also manage appointments and schedules.

Medical Record Management: Doctors can add diagnosis information, treatment details, and prescriptions. Patients' medical history will be stored securely.

Laboratory Management: The system will manage laboratory test requests and store test results for future reference.

Billing Management: The system will generate bills based on hospital services and maintain payment records.

Reporting System: The system will generate different reports for hospital administrators to monitor activities and performance.

1.3 Objectives:

The main objectives of the National Hospital Management System are:

- To reduce manual hospital management processes.
- To maintain patient information digitally.

- To provide quick access to medical records.
- To improve appointment scheduling efficiency.
- To reduce errors in billing and record management.
- To improve communication between patients and healthcare providers.
- To provide secure storage of healthcare data.

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
NHMS	National Hospital Management System
SRS	Software Requirements Specification
EMR	Electronic Medical Record
UI	User Interface
DBMS	Database Management System
Admin	System Administrator
OPD	Outpatient Department
API	Application Programming Interface

1.5 Intended Audience:

This SRS document is intended for : developers, testers, hospital administrators, and academic evaluators.

Chapter 2: Overall Description

2.1 Product Perspective:

National Hospital Management System (NHMS) is a web-based system designed to automate hospital operations. It provides a centralized platform for managing patient information, doctor records, appointments, medical records, laboratory reports, and billing activities.

2.2 Product Functions:

The major functions of the system are:

- User Authentication
- Patient Management
- Doctor Management
- Appointment Management
- Medical Record Management
- Prescription Management
- Laboratory Management
- Billing Management
- Report Generation

2.3 User Characteristics:

User	Responsibilities
Admin	Manage system and users
Receptionist	Register patients and manage appointments
Doctor	Manage diagnosis and prescriptions
Lab Technician	Manage laboratory reports
Patient	View appointments and reports

2.4 Operating Environment:

Hardware Requirements:

- Processor: Intel Core i3 or higher

- RAM: 4 GB or higher
- Storage: 100 GB

Software Requirements:

- Operating System: Windows 10 or above
- Web Browser: Chrome, Firefox, Edge
- Database: MySQL
- Server: XAMPP

2.5 Assumptions and Dependencies:

- Internet connection is available.
- Users have basic computer knowledge.
- Database server remains operational.
- Authorized users maintain login credentials securely.

2.6 Constraints:

- Limited project development time.
- Budget limitations.
- Data security requirements.
- Dependence on network availability.

Chapter 3: Functional Requirements and Use Case Specifications

3. 1 Functional Requirements:

FR-01: User Authentication: The system shall allow authorized users to log in and access the system according to their assigned roles.

Actors: Admin, Doctor, Receptionist, Laboratory Staff, Patient

Priority: High

FR-02: Patient Management: The system shall allow users to register, update, search, and manage patient information.

Actors: Admin, Receptionist

Priority: High

FR-03: Doctor Management: The system shall allow administrators to add, update, and manage doctor information and schedules.

Actors: Admin

Priority: High

FR-04: Appointment Management: The system shall allow appointment scheduling, updating, and cancellation between patients and doctors.

Actors: Receptionist, Patient, **Priority:** High

FR-05: Medical Record Management: The system shall store and maintain patient medical records, diagnosis details, and treatment history.

Actors: Doctor

Priority: High

FR-06: Prescription Management: The system shall allow doctors to create and manage patient prescriptions.

Actors: Doctor

Priority: Medium

FR-07: Laboratory Management: The system shall manage laboratory test requests, results, and reports.

Actors: Laboratory Staff

Priority: Medium

FR-08: Billing Management: The system shall generate bills and maintain payment records for hospital services.

Actors: Admin, Receptionist

Priority: Medium

FR-09: Report Generation: The system shall generate reports related to patients, appointments, billing, and hospital activities.

Actors: Admin, **Priority:** Medium

3.2 Use Case Descriptions:

UC-01: Login

Item	Description
Actor	Admin, Doctor, Receptionist, Laboratory Staff, Patient
Description	User logs into the system using valid credentials.
Precondition	User account exists.
Main Flow	Enter username and password → Click Login.
Postcondition	User dashboard is displayed.

UC-02: Register Patient

Item	Description
Actor	Receptionist
Description	Register a new patient in the system.
Precondition	Receptionist is logged in.
Main Flow	Enter patient information → Save record.
Postcondition	Patient profile is created successfully.

UC-03: Book Appointment

Item	Description
Actor	Receptionist
Description	Schedule an appointment for a patient.

Precondition	Patient and doctor records exist.
Main Flow	Select patient → Select doctor → Choose appointment date.
Postcondition	Appointment is successfully scheduled.

UC-04: Update Medical Record

Item	Description
Actor	Doctor
Description	Record diagnosis and treatment information.
Precondition	Patient record exists.
Main Flow	Open patient record → Enter diagnosis → Save information.
Postcondition	Medical record is updated successfully.

UC-05: Generate Bill

Item	Description
Actor	Admin, Receptionist
Description	Generate a bill for hospital services.
Precondition	Service information is available.
Main Flow	Enter charges → Generate bill.
Postcondition	Bill is created and stored successfully.

Chapter 4: Non-Functional Requirements

4.1 Performance Requirements:

- The system shall respond to user requests within 3 seconds under normal conditions.
- The system shall support multiple users accessing the system simultaneously.
- Patient records and reports shall be retrieved efficiently.

4.2 Security Requirements:

- Only authorized users shall access the system.
- Users shall log in using a valid username and password.
- Role-based access control shall be implemented.
- Patient information shall be protected from unauthorized access.

4.3 Usability Requirements:

- The system shall provide a simple and user-friendly interface.
- Users shall be able to navigate the system easily.
- Forms and menus shall be clear and understandable.

4.4 Reliability Requirements:

- The system shall maintain data accuracy and consistency.
- The database shall store records securely.
- The system shall minimize data loss during operation.

4.5 Scalability Requirements:

- The system shall support an increasing number of patients and users.
- Additional hospitals and departments can be added in the future without major system changes.

4.6 Maintainability Requirements:

- The system shall allow easy modification and updates.
- Errors and issues shall be easy to identify and fix.

- New features can be integrated with minimal impact on existing functions.

Chapter 5: External Interface Requirements

5.1 User Interface: The system shall provide a web-based user interface for all users.

Major interfaces include:

- Login Page
- Dashboard
- Patient Management Page
- Appointment Management Page
- Medical Record Page
- Billing Page
- Report Page

5.2 Hardware Interface:

- Computer or Laptop
- Minimum 4 GB RAM
- Internet Connection

5.3 Software Interface: The system shall use the following software components:

- Operating System: Windows 10 or above
- Web Browser: Google Chrome, Mozilla Firefox
- Web Server: Apache (XAMPP)
- Database: MySQL
- Programming Language: PHP

5.4 Database Interface: The system shall interact with a MySQL database to store and retrieve information related to:

- Patients
- Doctors

- Appointments
- Medical Records
- Laboratory Reports
- Billing Information

Chapter 6: Agile Requirement Engineering

6.1 User Stories:

US-01: As a patient, I want to book an appointment so that I can consult a doctor easily.

US-02: As a doctor, I want to view patient medical records so that I can provide proper treatment.

US-03: As a receptionist, I want to register patients so that patient information can be stored efficiently.

US-04: As an administrator, I want to generate reports so that hospital activities can be monitored.

6.2 Acceptance Criteria:

AC-01 Appointment Booking

- Patient selects a doctor and appointment date.
- System confirms the appointment successfully.

AC-02 Patient Registration

- Receptionist enters valid patient information.
- System creates a patient record successfully.

AC-03 Medical Record Management

- Doctor enters diagnosis and treatment details.
- System stores the information successfully.

AC-04 Report Generation

- Administrator selects a report type.
- System generates the requested report.

6.3 Product Backlog

Priority	Backlog Item
High	User Authentication
High	Patient Management
High	Doctor Management
High	Appointment Management
Medium	Medical Record Management
Medium	Laboratory Management
Medium	Billing Management
Low	Report Generation

Chapter 7: Database Requirements

7.1 Database Overview: The National Hospital Management System (NHMS) will use a MySQL database to store and manage hospital information. The database will maintain records related to patients, doctors, appointments, medical records, prescriptions, laboratory reports, billing information, and user accounts.

7.2 Patient Table

Field Name	Data Type	Description
patient_id	INT (PK)	Patient ID
name	VARCHAR(100)	Patient name
age	INT	Age
gender	VARCHAR(10)	Gender
phone	VARCHAR(20)	Phone number
address	VARCHAR(255)	Address

7.3 Doctor Table

Field Name	Data Type	Description
doctor_id	INT (PK)	Doctor ID
doctor_name	VARCHAR(100)	Doctor name
specialization	VARCHAR(100)	Specialization
phone	VARCHAR(20)	Phone number
schedule	VARCHAR(100)	Schedule

7.4 Appointment Table

Field Name	Data Type	Description
appointment_id	INT (PK)	Appointment ID
patient_id	INT (FK)	Patient ID
doctor_id	INT (FK)	Doctor ID
appointment_date	DATE	Appointment date
status	VARCHAR(20)	Status

7.5 Medical Record Table

Field Name	Data Type	Description
record_id	INT (PK)	Record ID
patient_id	INT (FK)	Patient ID
doctor_id	INT (FK)	Doctor ID
diagnosis	TEXT	Diagnosis
treatment	TEXT	Treatment
visit_date	DATE	Visit date

7.6 Prescription Table

Field Name	Data Type	Description
prescription_id	INT (PK)	Prescription ID
patient_id	INT (FK)	Patient ID
doctor_id	INT (FK)	Doctor ID
medicine	TEXT	Medicine
dosage	VARCHAR(100)	Dosage

7.7 Laboratory Test Table

Field Name	Data Type	Description
test_id	INT (PK)	Test ID
patient_id	INT (FK)	Patient ID

Field Name	Data Type	Description
test_name	VARCHAR(100)	Test name
test_result	TEXT	Test result
test_date	DATE	Test date

7.8 Billing Table

Field Name	Data Type	Description
bill_id	INT (PK)	Bill ID
patient_id	INT (FK)	Patient ID
amount	DECIMAL(10,2)	Amount
payment_status	VARCHAR(20)	Payment status
bill_date	DATE	Bill date

Chapter 8: System Models and Diagrams

8.1 Use Case Diagram:

The Use Case Diagram illustrates the interactions between users and the National Hospital Management System (NHMS). It shows how different users access and utilize the system functionalities according to their roles.

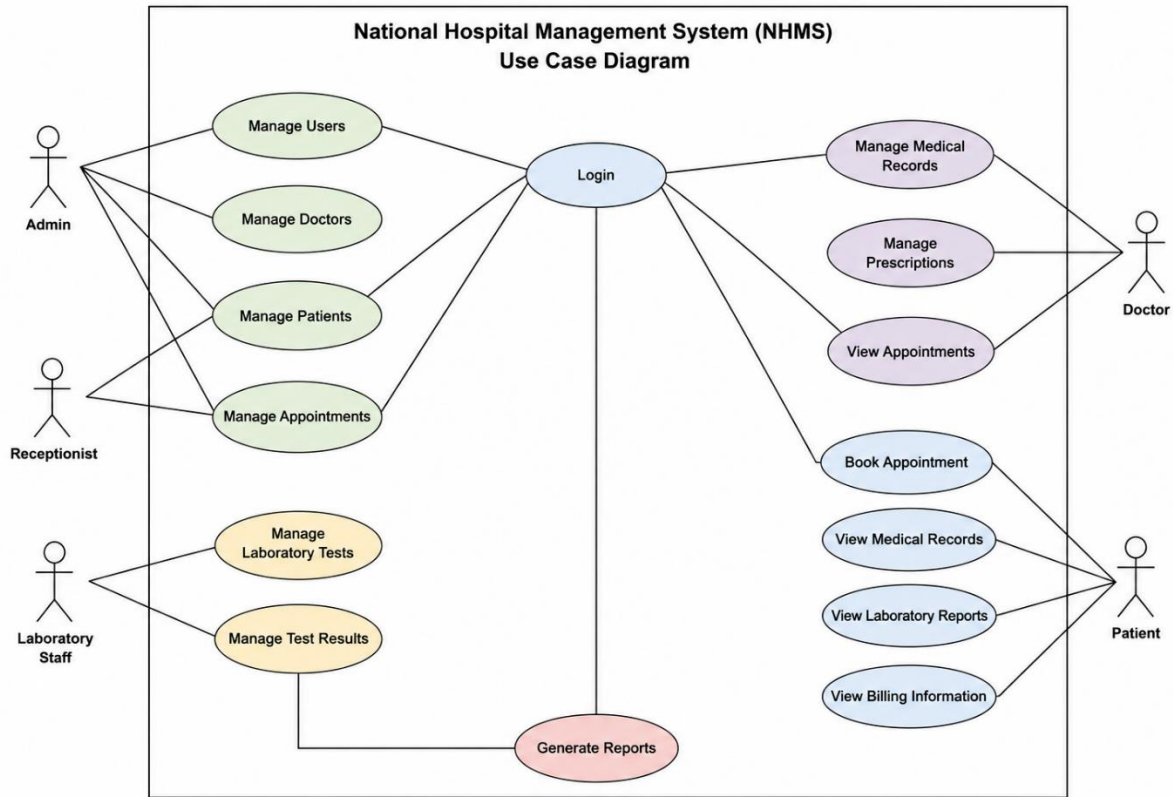


Fig.NO.: 01: Use Case Diagram (NHMS)

8.2 Context Diagram (DFD Level 0):

The Context Diagram provides a high-level view of the NHMS. It shows the interaction between the system and external entities such as Admin, Receptionist, Doctor, Laboratory Staff, and Patient.

NHMS - Context Diagram (DFD Level 0)

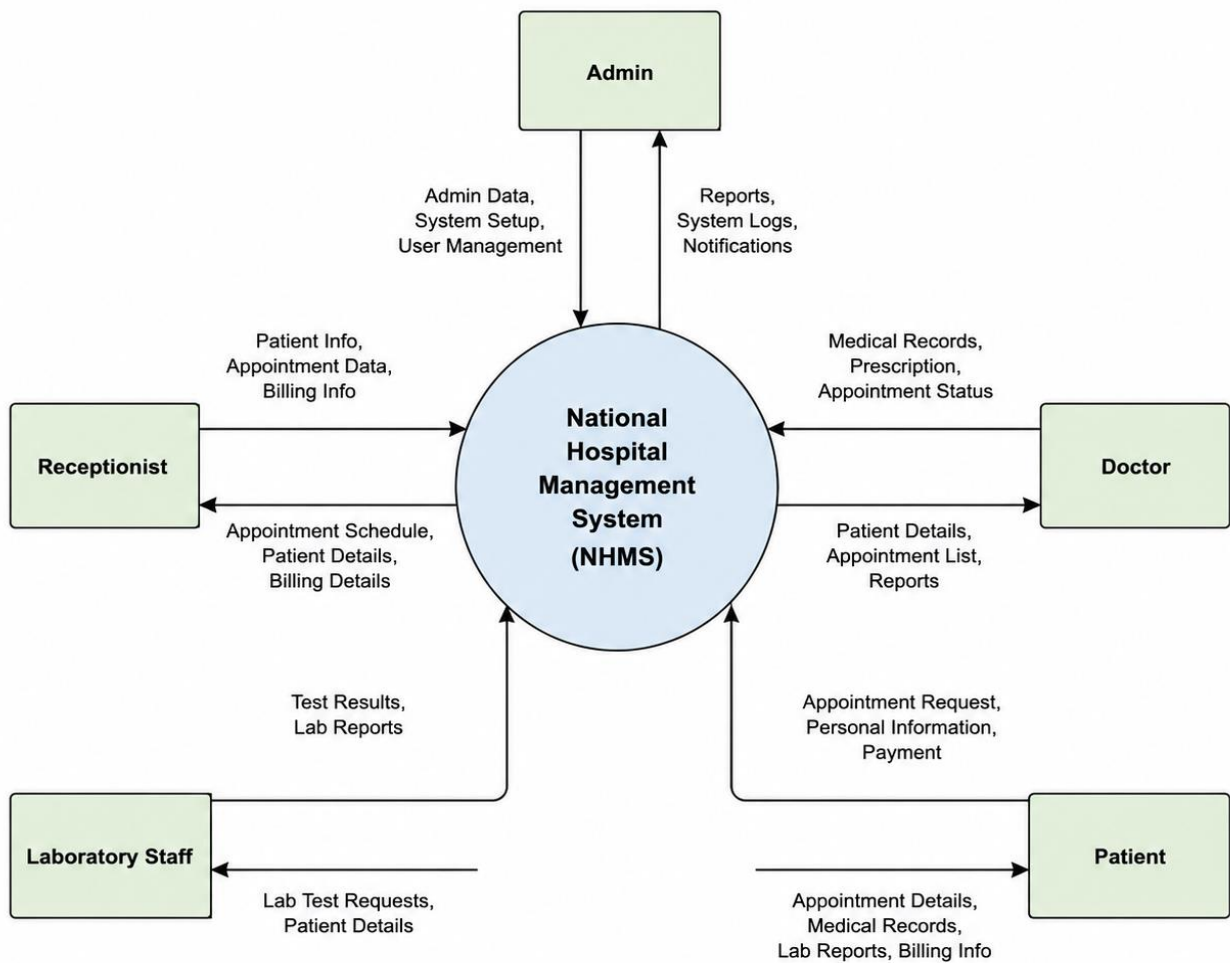


Fig.NO: 02: Context Diagram (DFD Level 0)-NHMS

8.3 DFD Level 1:

The Level 1 Data Flow Diagram illustrates the major processes of the system, including patient management, doctor management, appointment management, medical records, laboratory services, billing, and report generation.

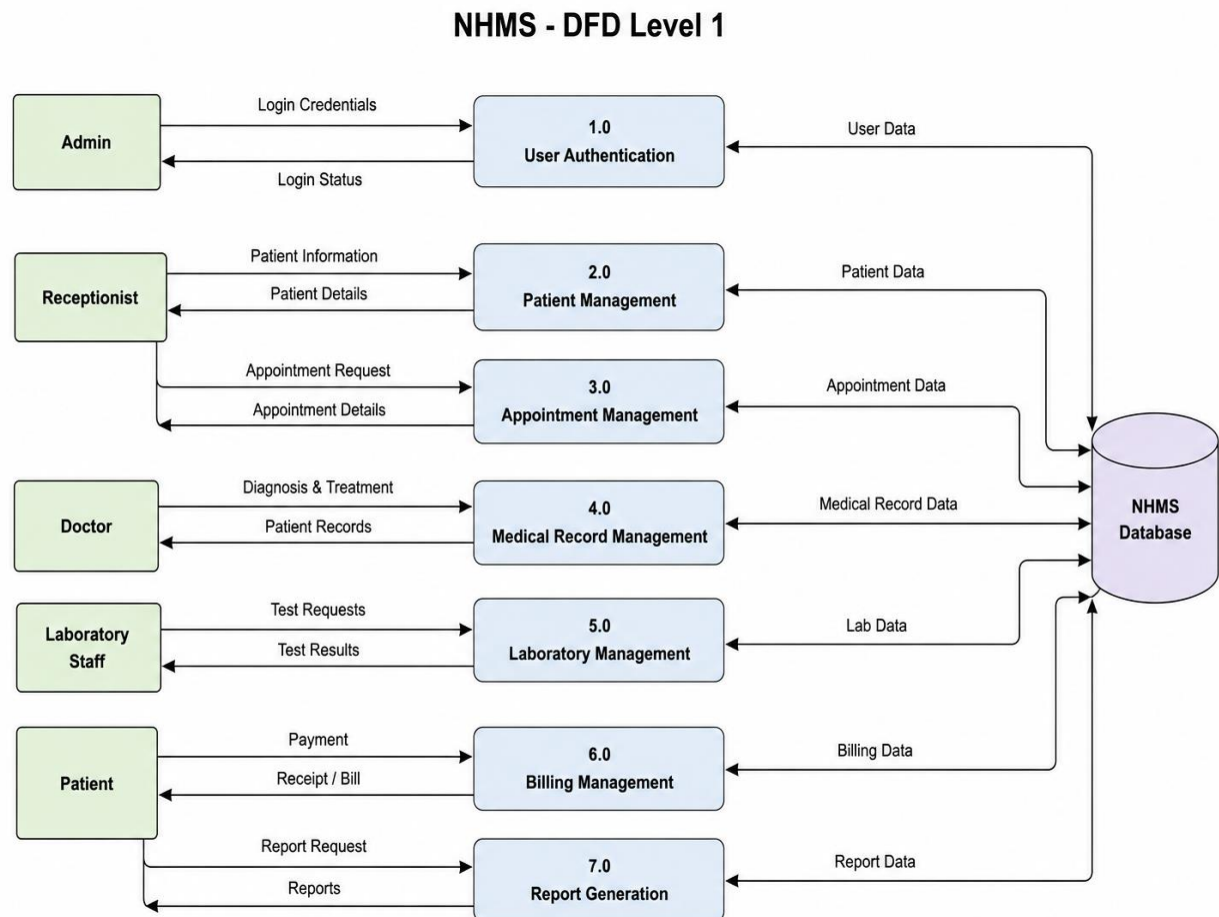


Fig.NO: 03: DFD Level 1 (NHMS)

8.4 Entity Relationship Diagram (ERD):

The Entity Relationship Diagram represents the database structure of NHMS and shows the relationships among patients, doctors, appointments, medical records, prescriptions, laboratory tests, and billing information.

NHMS - Entity Relationship Diagram (ERD)

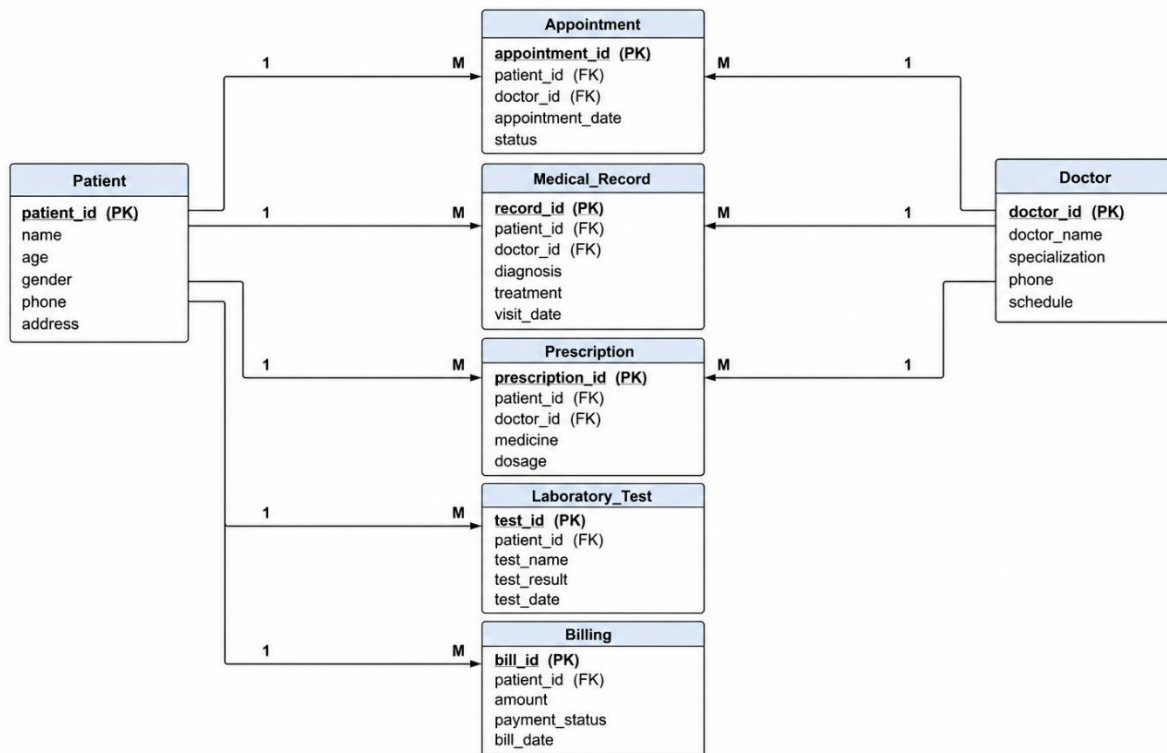


Fig.NO: 04: Entity Relationship Diagram (ERD)-NHMS

Chapter 9: Future Enhancement

The National Hospital Management System (NHMS) can be further enhanced by incorporating additional features in future versions. Potential improvements include:

1. Online patient registration and appointment booking.
2. Mobile application support for patients and doctors.
3. SMS and email notification services.
4. Online payment integration.
5. Advanced reporting and analytics features.
6. Integration with national healthcare databases.
7. Telemedicine and virtual consultation support.

Chapter 10: Conclusion

The National Hospital Management System (**NHMS**) is designed to provide an efficient and centralized solution for managing hospital operations. The system supports patient management, doctor management, appointments, medical records, laboratory services, and billing processes. By automating these activities, NHMS can improve operational efficiency, data accuracy, and service quality. This SRS document serves as a foundation for the successful development and implementation of the proposed system.